

Department of Statistics and Computer Science
University of Peradeniya
CSC2021 - Programming using Data Structures
Labsheet 01
INTRODUCTION TO DATA STRUCTURES, ARRAYS

Purpose

- Introduce the basic operations on data structures using an Array
- Introduce the basic composition of advanced Data Structures.

Theory

- A Data structure is a set of elements of a certain type.
- The basic operations on data structures can be listed as follows,
 - Insertion
 - Search
 - Modification
 - Deletion
 - Retrieval
- An array is the most simple form of a data structure.
- It is a static data structure (i.e. The size of an Array is stated at the point of declaration.)

Task 1

Implement a java program to reverse the elements of the array given below:

double datapoints[] = {2.5,19.7,8.95,11.0,28.4,7.11,4.16};

Task 2

Write a program to multiply the following matrices as given below:

$$MatrixA = \begin{bmatrix} 2 & 5 & 8 \\ 4 & 8 & 3 \\ 9 & 1 & 6 \end{bmatrix}$$

$$MatrixB = \begin{bmatrix} 1 & 7 & 9 \\ 6 & 4 & 5 \\ 2 & 8 & 3 \end{bmatrix}$$

$$MatrixA * MatrixB = \begin{bmatrix} (2*1+5*6+8*2) & (2*7+5*4+8*8) & (2*9+5*5+8*3) \\ (4*1+8*6+3*2) & (4*7+8*4+3*8) & (4*9+8*5+3*3) \\ (9*1+1*6+6*2) & (9*7+1*4+6*8) & (9*9+1*5+6*3) \end{bmatrix}$$

$$MatrixA * MatrixB = \begin{bmatrix} 48 & 98 & 71 \\ 58 & 84 & 75 \\ 27 & 111 & 104 \end{bmatrix}$$

Task 3

Eight leading companies decide to invest in different shares as shareholders of the share market and each shareholder name and the price of the shares they have invested in is given below:

Shareholder Name	Shareholder Amount
Cargills	100000
Union Bank	300000
Laugh Gas	200000
Odel	400000
Melstacorp	600000
Kingsbury	150000
Hemas	250000
Brandix	450000

Due to the pandemic situation one of the companies named **Melstacorp** decided to pull out from their invested shares as the company was going through a very lean period. Write a java program to insert the above shareholders into an array and remove the appropriate shareholder as given above.

The sharemarket wanted an annual report of the remaining set of companies containing the Shareholder name and Shareholder amount sorted in ascending order of the shareholder's amount. Implement a suitable method to carry out the requirement